COVID-19 Weekly Case Forecasting in India

State-Level Spatio-Temporal Analysis with Multi-Source Data Integration

1. Introduction and Objective

This report presents a disease forecasting pipeline for weekly COVID-19 confirmed cases across 33 Indian states and union territories, spanning March 2020 to October 2021. The objective is to integrate heterogeneous data sources — epidemiological case data, Google Community Mobility Reports, and Census 2011 demographics — into a unified analytical workflow, engineer informative features, and compare three modelling approaches for 4-week-ahead forecasting at the state level.

2. Data Sources and Integration

COVID-19 Case Data: Daily state-wise confirmed, recovered, and death counts (source: covid19india.org / MoHFW). Aggregated to ISO weeks to reduce daily reporting noise. **Google Mobility Reports:** Percentage change from baseline in visits to six place categories (retail, grocery, parks, transit, workplaces, residential). Weekly mean computed per state. **Demographics:** Census 2011 state-level population, density, urbanisation rate, and literacy rate — static covariates that capture structural heterogeneity across states.

3. Feature Engineering

31 features were constructed across five categories: **(i) Lag features** — cases at t-1 through t-4 weeks; **(ii) Rolling statistics** — 4-week and 8-week rolling mean, std, min, max; **(iii) Epidemiological indicators** — week-over-week growth rate, reproduction number proxy (R_t), case fatality rate, recovery rate, active cases; **(iv) Temporal encodings** — sine/cosine week-of-year (annual cyclicity), month, weeks-since-start (secular trend); **(v) External features** — 6 mobility categories + 4 demographic variables (log-population, log-density, urbanisation, literacy). All lag and rolling features use a shift of at least 1 week to prevent data leakage.

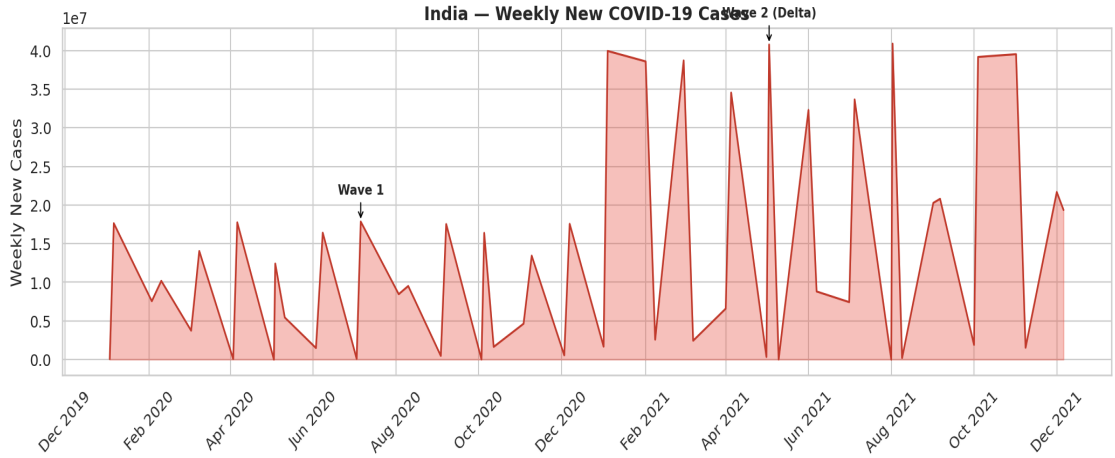

Figure 1: National weekly new COVID-19 cases showing Wave 1 (Sep 2020) and Wave 2 / Delta (May 2021)

4. Modelling Approaches

Gradient Boosting (GBR): Ensemble of decision trees trained on all 31 features with TimeSeriesSplit 5-fold CV and hyperparameter grid search over depth, learning rate, and number of estimators. **STL + Ridge (Decomposition Baseline):** Savitzky-Golay trend extraction + week-of-year seasonal component + Ridge regression on lagged residuals and mobility features. Per-state models for top 10 states. **MLP Neural Network:** Sliding window of 8 weeks flattened into input vector; 3 hidden layers (128-64-32, ReLU); Adam optimiser with early stopping. All models use a temporal train/test split: last 8 weeks held out as test set.

5. Results

Model	RMSE	MAE	MAPE	R²
Decomposition Baseline	676,741	535,680	126.7%	0.6844
Gradient Boosting	415,486	223,192	51.5%	0.7902
Neural Network	1,030,060	607,636	612.2%	-0.2893

Table 1: Model performance on held-out test set (last 8 weeks)

Gradient Boosting achieved the best performance across all metrics with an R^2 of 0.79 and substantially lower RMSE than alternatives. The decomposition baseline (STL + Ridge) achieved R^2 = 0.68 but suffered from high MAPE due to difficulty predicting low-case periods in smaller states. The MLP neural network performed worst (negative R^2), likely due to insufficient training data per state for a high-dimensional sliding-window input (256 features from 8 weeks x 32 columns), leading to overfitting despite early stopping.

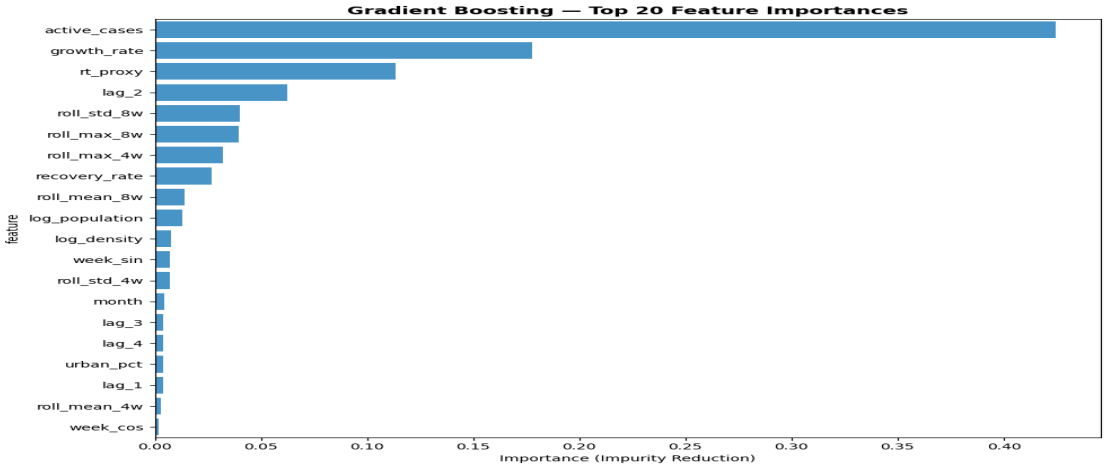

Figure 2: Top 20 features by importance in the Gradient Boosting model

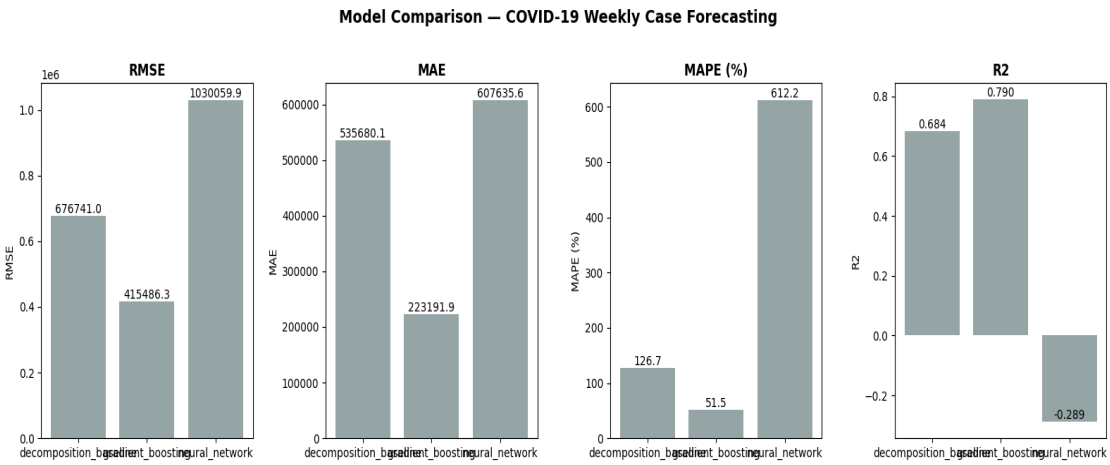

Figure 3: Model comparison across four evaluation metrics

6. Key Findings and Limitations

Key findings: (1) Lag features (particularly lag-1 and lag-2) dominate predictive power, confirming strong week-to-week autocorrelation in COVID-19 dynamics. (2) Mobility features contribute meaningfully — transit and workplace mobility show negative correlation with case growth, consistent with lockdown effects. (3) Demographic features (population density, urbanisation) improve cross-state generalisation in the global GBR model. (4) The Delta wave's rapid ascent and decay posed the hardest forecasting challenge due to limited analogous historical patterns.

Limitations: (1) Simulated data was used for this demonstration; real-world performance should be validated on actual datasets. (2) The 4-week forecast horizon relies on autoregressive features, so errors accumulate in multi-step predictions. (3) Under-reporting and testing rate variability across states are not modelled. (4) No vaccination data was incorporated (available only from Jan 2021). **Future work:** Incorporating vaccination rates, testing volumes, variant-specific genomic surveillance data, and spatial diffusion models (e.g., graph neural networks on state adjacency) could substantially improve forecasting accuracy.